Rakamin Academy × Bank Muamalat

Project Based Internship **Apr 2026**

SALES PERFORMANCE DASHBOARD

PT SEJAHTERA BERSAMA

Business Intelligence Analyst

Presented By: Jonathan Marjono

KEY TOPICS

01 Introduction

02 Case Study

03 Tools Used

04 Dataset

05 ERD

06 Master Table

07 Dasboard Creation

08 Exploratory Data Analysis

09 Key Insights

10 Recommendations

INTRODUCTION

Jonathan Marjono

About Me

A fresh graduate in Statistics with a strong foundation in data analysis and a growing interest in becoming a Data and Business Analyst. I am data-driven and enthusiastic about working with data to uncover insights and support decision-making. I am also able to work collaboratively, both independently and as part of a team, to deliver effective solutions.

Experience

Data & Operations Analyst Intern

Allianz Life Indonesia | Premium Management
Jul 2025 - Jan 2026

Data Analyst Intern

Bank Rakyat Indonesia | Regional Decision Support
Aug 2024 - Jan 2025

Formal Education

IPB University

Statistics & Data Science Study Program
Aug 2021 - Jul 2025

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Apr 2026

Internship

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CASE STUDY

As a **Business Intelligence** at **Bank Muamalat**, I developed a dashboard for **PT Sejahtera Bersama** using **four** separate sales **datasets**. The process began by **identifying the primary key** in each table and **determining the relationships** between them to ensure accurate data integration and data consistency.

Next, selected components from each dataset were **combined into a single master table**. This master table served as the foundation for **building visualizations and an interactive dashboard**, enabling efficient **data exploration**.

The dashboard was designed to generate insights that support **data-driven decision-making**. It helps identify sales trends, evaluate performance, and **provide actionable business recommendations** to maintain or improve sales growth.

TOOLS USED

Google Bigquery

Used to import datasets, combine multiple tables, and perform data manipulation using SQL for data processing and analysis

Google Sheets

Used to read and review tabular data, convert files between .xlsx and .csv formats, and store datasets. Also used to save processed data from BigQuery and connect it to Data Studio for dashboard creation.

Data Studio (Looker Studio)

Used to create visualizations and interactive dashboard

Eraser.io

Used to create Entity Relationship Diagrams (ERD) to visualize table structures and relationships between datasets

03

04

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#1 Customers

```
SELECT *  FROM `rakamin-493213.UserChurn.Customers` LIMIT 10;
```

- CustomerID : unique customer identifier
- FirstName : customer's first name
- LastName : customer's last name
- CustomerEmail : customer's email address
- CustomerPhone : customer's phone number
- CustomerAddress : customer's address
- CustomerCity : customer's city
- CustomerState : customer's state
- CustomerZip : customer's postal code

Row	CustomerID	FirstName	LastName	CustomerEmail	CustomerPhone	CustomerAddress	CustomerCity	CustomerState	CustomerZip
1	1368	Buck	Meiklam	bmeiklamiv@myspace.com#m...	205-789-4928	8943 Rusk Drive	Birmingham	Alabama	35205
2	260	Charlena	Lille	clillea8@nasa.gov#mailto:clille...	205-464-9921	13293 Macpherson Pass	Birmingham	Alabama	35215
3	1694	Robinia	Balog	rbalogiw@arstechnica.com#ma...	205-133-6098	565 Fairfield Terrace	Birmingham	Alabama	35220
4	1480	Bryna	Cumberpatch	bcumberpatchjr@auda.org.au #mailto:bcumberpatchjr@auda.org.au#	205-731-4813	6665 Marcy Street	Birmingham	Alabama	35225
5	244	Jobie	Pinchen	jpinchen15@behance.net#mailt...	205-844-2402	4319 Coleman Lane	Birmingham	Alabama	35231
6	705	Esmeralda	McRory	emcrory6y@mac.com#mailto:e...	205-885-5499	368 Briar Crest Circle	Birmingham	Alabama	35242
7	954	Llewellyn	Fromont	lfromonte9@de.vu#mailto:lfro...	205-279-7028	14 Rowland Lane	Birmingham	Alabama	35244
8	1196	Betsy	Buncombe	bbuncombeqn@goodreads.co...	205-757-8485	30136 Fairview Hill	Birmingham	Alabama	35244
9	1360	Forrest	Lowdeane	flowdeaneb7@google.ca#mailt...	205-191-4282	728 Barnett Place	Birmingham	Alabama	35244
10	413	Demetrius	O' Liddy	dofz@ftc.gov#mailto:dofz@ftc....	205-983-4144	771 American Lane	Birmingham	Alabama	35254

DATASET

04

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#2 Products

```
SELECT *  FROM `rakamin-493213.UserChurn.Products` LIMIT 10;
```

- ProdNumber : unique product identifier
- ProdName : product name
- Category : product category ID
- Price : unit price of the product

Row	ProdNumber	ProdName	Category	Price
1	BP104	Cat Robot Blueprint	1	4.99
2	BP108	Panda Robot Blueprint	1	7.99
3	BP102	Bsquare Robot Blueprint	1	8.99
4	BP106	Hexacopter Drone Blueprint	1	8.99
5	BP101	All Eyes Drone Blueprint	1	9.99
6	BP109	QuadroCopter Blueprint	1	10.99
7	BP110	Sleepy Eye Blueprint	1	11.99
8	BP105	Creature Robot Arms Blueprint	1	12.0
9	BP107	Ladybug Robot Blueprint	1	12.0
10	BP111	Upside Down Robot Blueprint	1	12.0

DATASET

04

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#3 Orders

```
SELECT *  FROM `rakamin-493213.UserChurn.Orders` LIMIT 10;
```

- OrderID : unique order identifier
- Date : order date
- CustomerID : customer identifier
- ProdNumber : product identifier
- Quantity : quantity purchased

Row	OrderID	Date	CustomerID	ProdNumber	Quantity
1	1758	2021-01-15	923	BP101	1
2	404	2020-03-26	1256	BP101	2
3	575	2020-05-06	635	BP101	2
4	1098	2020-08-25	1480	BP101	2
5	1320	2020-10-10	534	BP101	2
6	1755	2021-01-14	1564	BP101	2
7	1756	2021-01-14	619	BP101	2
8	2029	2021-03-11	827	BP101	2
9	2870	2021-09-07	500	BP101	2
10	3158	2021-11-21	379	BP101	2

DATASET

04

#4 Product Category

```
SELECT *  FROM `rakamin-493213.UserChurn.Orders` LIMIT 10;
```

- CategoryID : unique category identifier
- CategoryName : category name
- CategoryAbbreviation : abbreviation of the category name

Row	CategoryID	CategoryName	CategoryAbbreviation
1	1	Blueprints	BP
2	2	Drone Kits	DK
3	3	Drones	DS
4	4	eBooks	EB
5	5	Robot Kits	RK
6	6	Robots	RS
7	7	Training Videos	TV

DATASET

ERD

```
Customers [icon: user, color: blue]{
  CustomerID INT64 pk(🔑)
  FirstName STRING
  LastName STRING
  CustomerEmail STRING
  CustomerPhone STRING
  CustomerAddress STRING
  CustomerCity STRING
  CustomerState STRING
  CustomerZip INT64
  createdAt timestamp
}

Orders [icon: shopping-cart, color: green] {
  OrderID INT64 pk(🔑)
  Date DATE
  CustomerID INT64 fk(🔗)
  ProdNumber STRING fk(🔗)
  Quantity INT64
  createdAt timestamp
}

Products [icon: box, color: orange] {
  ProdNumber STRING pk(🔑)
  ProdName STRING
  Category INT64 fk(🔗)
  Price FLOAT64
  createdAt timestamp
}

ProductCategory [icon: tag, color: purple] {
  CategoryID INT64 pk(🔑)
  CategoryName STRING
  CategoryAbbreviation STRING
  createdAt timestamp
}

Customers.CustomerID < Orders.CustomerID
ProductCategory.CategoryID < Products.Category
Products.ProdNumber < Orders.ProdNumber
```

Identification

- **Primary Key**: A column with unique, non-null values that distinctly represents each record in a table.
- **Foreign Key**: A column that references another table to establish relationships between datasets.

Primary Key

Customers: CustomerID
Orders: OrderID
Products: ProdNumber
ProductCategory: CategoryID

Foreign Key

Customers: -
Orders: CustomerID, ProdNumber
Products: Category
ProductCategory: -

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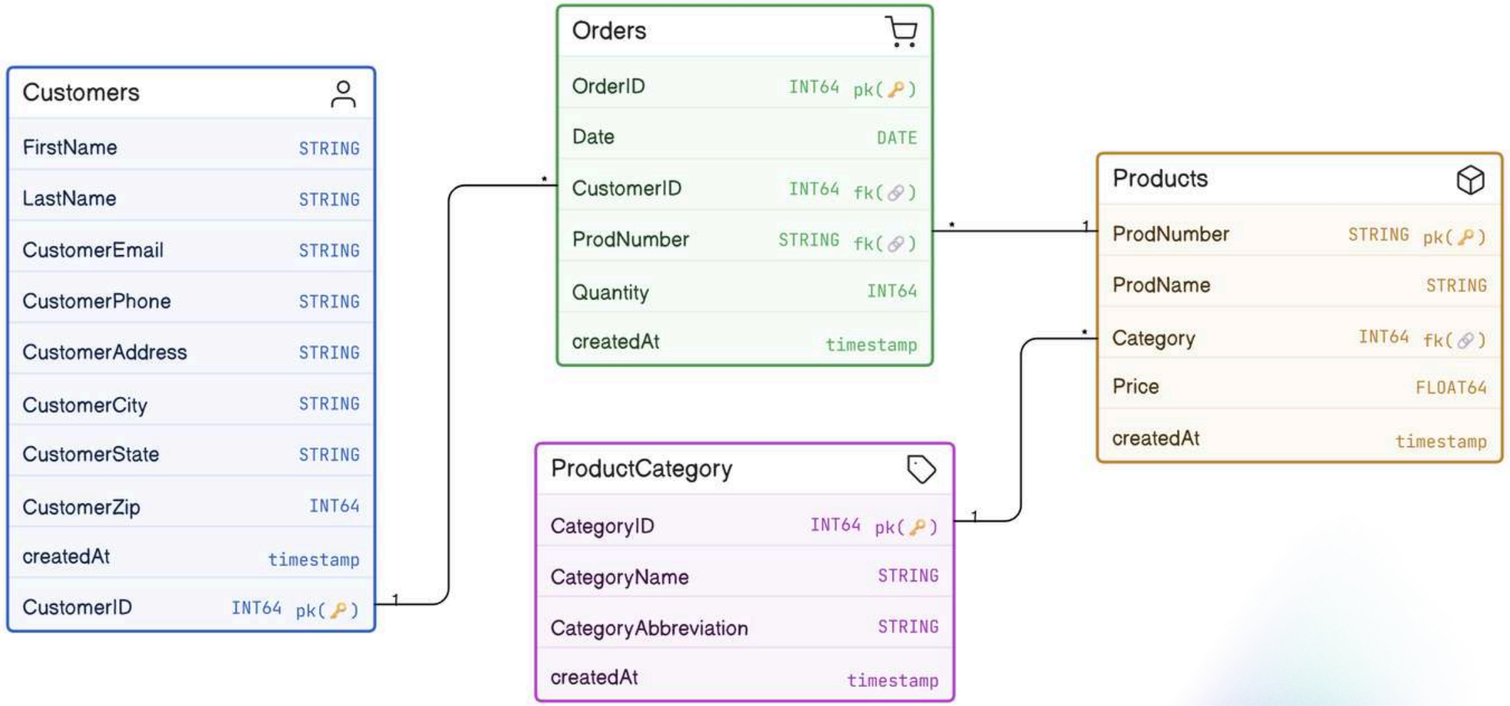
ERD



After identifying the **primary** and **foreign keys**, the relationships between tables can be defined by matching **foreign key columns** with their **corresponding primary keys** in other tables. This process determines how the tables are connected and forms relationships such as **one-to-one** (1:1), **one-to-many** (1:N), or **many-to-many** (N:M) based on **how the data is associated**.

In this project, the relationships between the four tables are structured as **one-to-many**, where a single record in a **parent table** can be associated with multiple records in a **child table** through **foreign keys**.

Customers < Orders: One to Many
(One customer can have multiple orders)

Products < Orders: One to Many
(One product can appear in many orders)

ProductCategory < Products: One to Many
(One product category contains many products)

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MASTER TABLE

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*Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule | Open in | More

```google_sql_query
WITH transaction_summary AS (
    SELECT
        o.Date                                          AS order_date,
        pc.CategoryName                                 AS category_name,
        p.ProdName                                      AS product_name,
        p.Price                                         AS product_price,
        o.Quantity                                      AS order_qty,
        ROUND(CAST(p.Price AS NUMERIC) * o.Quantity, 2) AS total_sales,
        SPLIT(c.CustomerEmail, '#')[OFFSET(0)]          AS cust_email,
        c.CustomerCity                                  AS cust_city

    FROM       `rakamin-493213.UserChurn.Orders`          o
    INNER JOIN `rakamin-493213.UserChurn.Products`        p  ON p.ProdNumber = o.ProdNumber
    INNER JOIN `rakamin-493213.UserChurn.ProductCategory` pc ON pc.CategoryID = p.Category
    INNER JOIN `rakamin-493213.UserChurn.Customers`       c  ON c.CustomerID  = o.CustomerID
)

SELECT *
FROM   transaction_summary
ORDER BY order_date ASC, order_qty ASC;
```

Common Table Expression (CTE)

total_sales = (price * quantity)

transaction summmary

inner join

retrieve all fields from transaction summary

order by order date & order quantity

Query completed

Using on-demand processing quota

Query results | Chat | Save results | Op

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	order_date	category_name	product_name	product_price	order_qty	total_sales	cust_email	cust_city
1	2020-01-01	Drone Kits	BYOD-220	69.0	1	69	edew@nba.com	Honolulu
2	2020-01-01	eBooks	Polar Robots	23.99	2	47.98	fvaslerqt@comsenz.com	Jackson
3	2020-01-01	Robots	RWW-75 Robot	883.0	3	2649	tmckernot@tinyurl.com	Katy
4	2020-01-01	eBooks	SCARA Robots	19.5	5	97.5	llespercx@com.com	Des Moines
5	2020-01-01	eBooks	Spherical Robots	16.75	5	83.75	lfromonte9@de.vu	Birmingham
6	2020-01-01	Training Videos	Drone Video Techniques	37.99	6	227.94	gstiggersdd@eventbrite.com	Saint Petersburg
7	2020-01-02	Training Videos	Understanding Automation	44.95	1	44.95	ksteershp@ameblo.jp	San Diego
8	2020-01-02	Blueprints	Ladybug Robot Blueprint	12.0	2	24	akingaby78@deviantart.com	West Palm Beach

Results per page: 50 | 1 – 50 of 3339

this is a preview of the transaction summary master table

Transaction Summary

- order_date
- category_name
- product_name
- product_price
- order_qty
- total_sales
- cust_email
- cust_city

DASHBOARD CREATION

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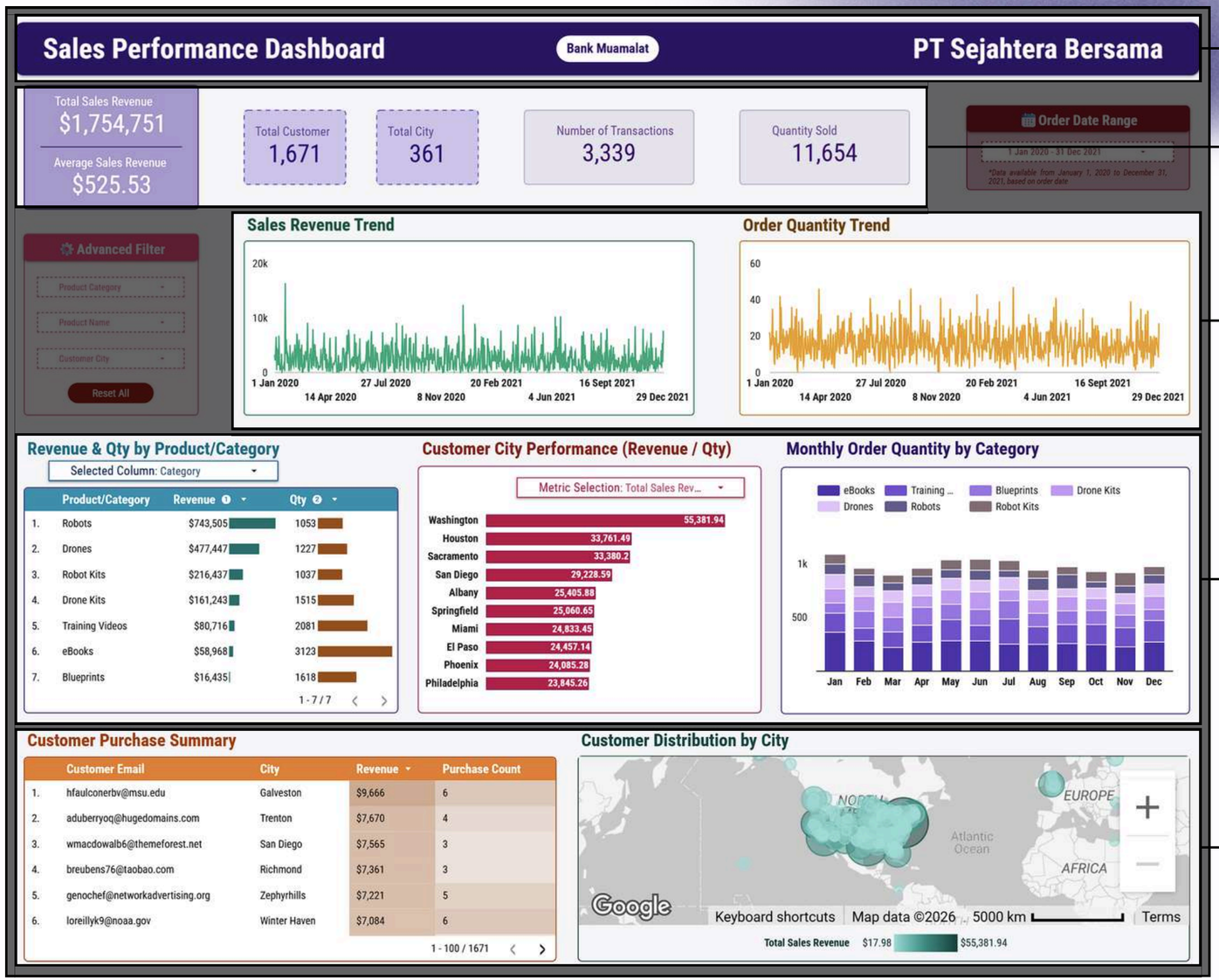

Header / Dashboard Title

Primary KPIs

Time-Series Trend Analysis

Product & Category Performance

Customer Profiles & Market Distribution

Sales Performance Dashboard PT Sejahtera Bersama

07

DASHBOARD CREATION

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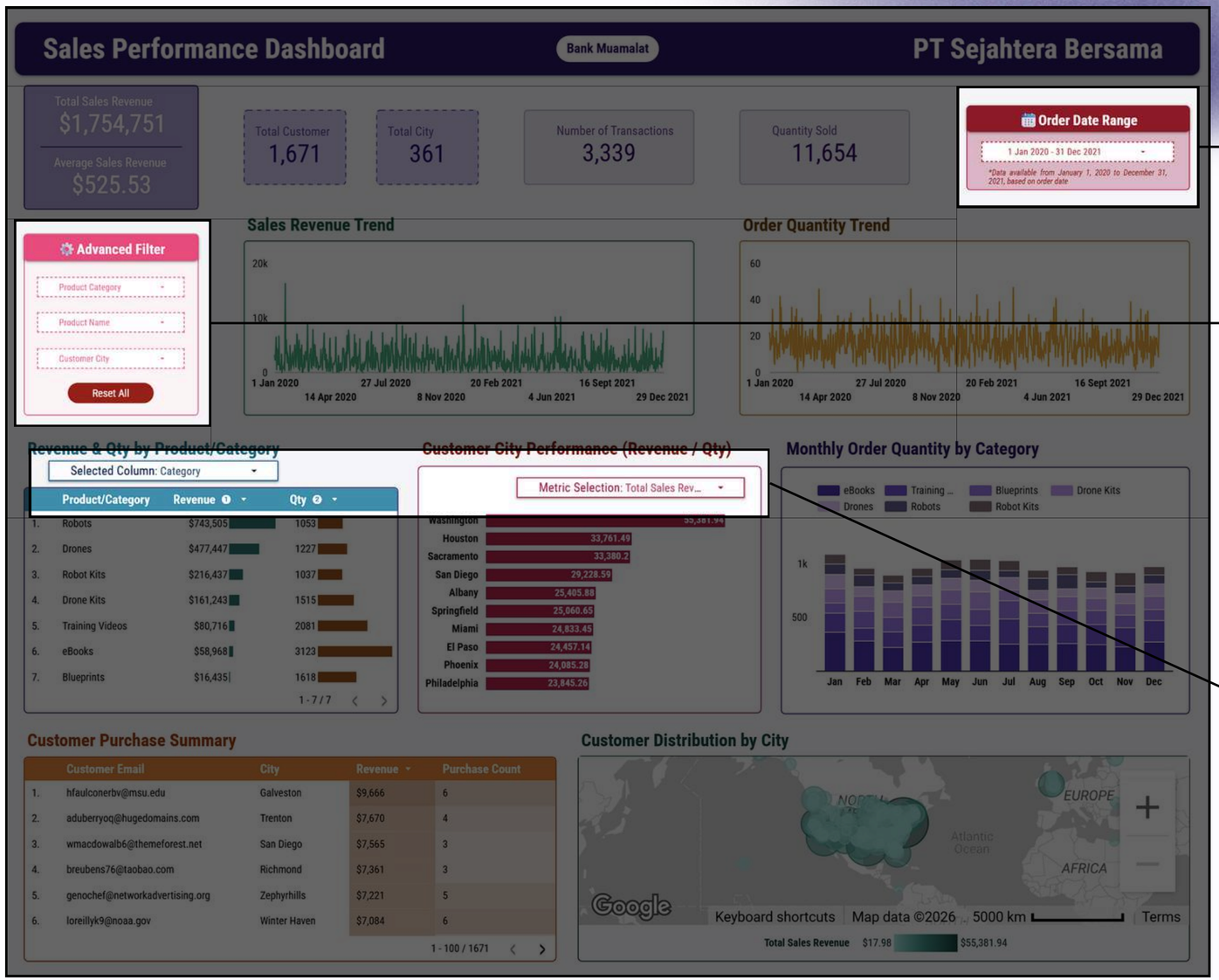
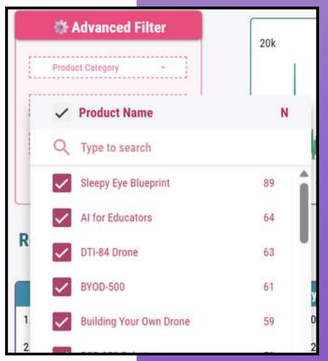
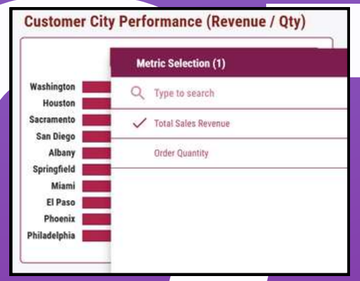


Date Basic Filter

Advanced Filtering & Navigation



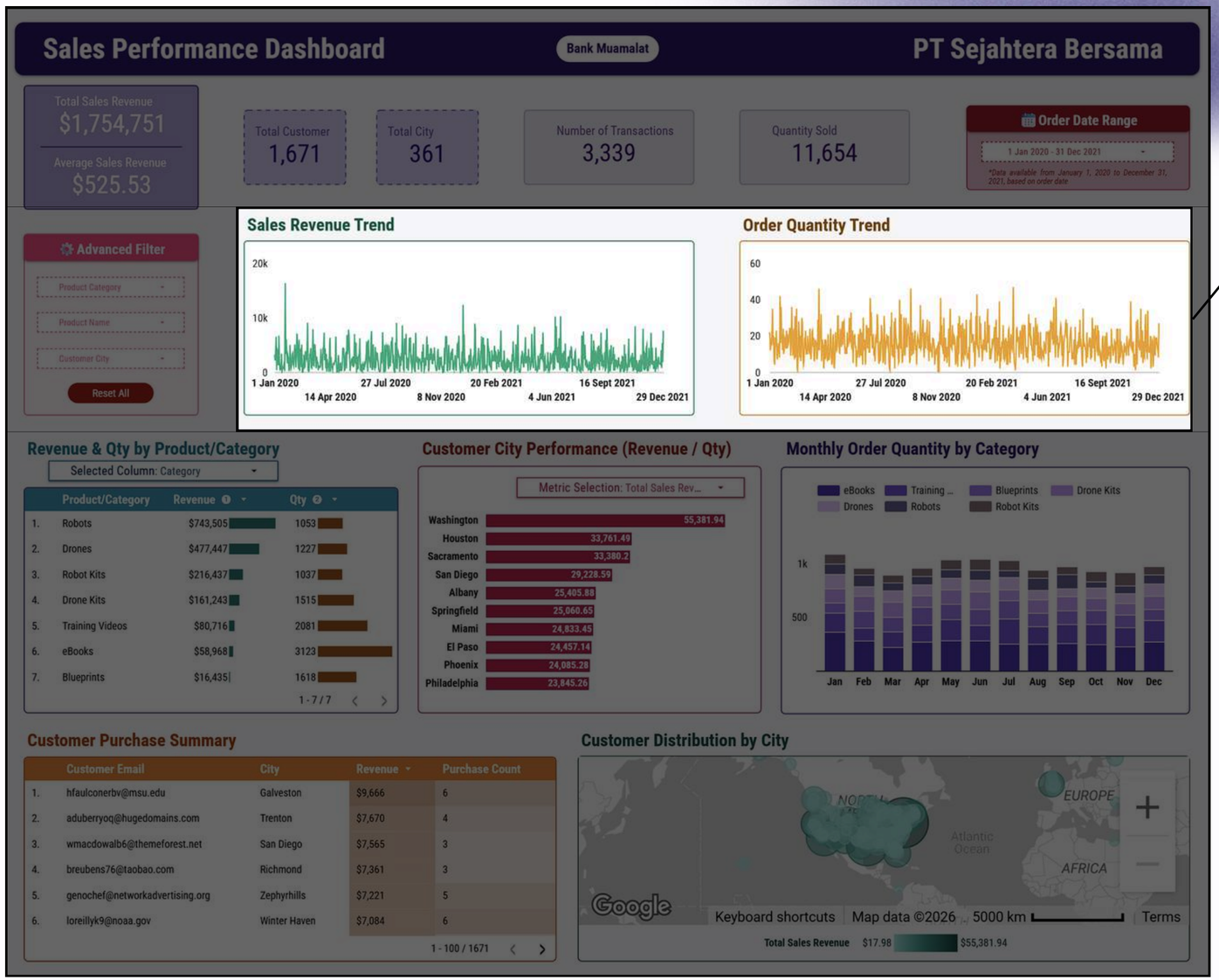
Dynamic Metric & Dimension Controls



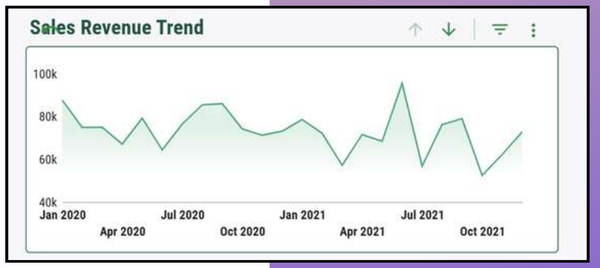
Sales Performance Dashboard PT Sejahtera Bersama

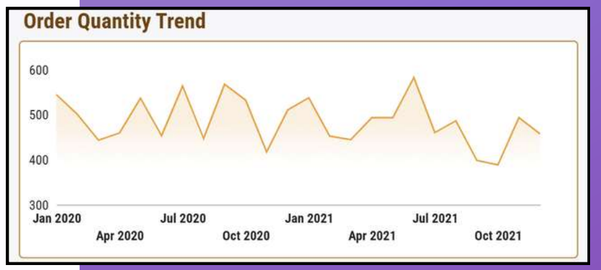
DASHBOARD CREATION

Rakamin Academy × Bank Muamalat



Multi-Level Time Granularity





07

[Sales Performance Dashboard PT Sejahtera Bersama]

EXPLORATORY DATA ANALYSIS

KPI Performance (2020-2021)

Total Sales Revenue	Total Customer	Total City	Number of Transactions	Quantity Sold
$1,754,751	1,671	361	3,339	11,654
Average Sales Revenue $525.53				

Total Sales

From 1 Jan 2020 to 31 Dec 2021, PT Sejahtera Bersama generated **$1,754,751** in sales within just two years, indicating **strong business performance**.

Total Customer

Within the period from 1 Jan 2020 to 31 Dec 2021, PT Sejahtera Bersama successfully acquired **1,671 unique customers**, indicating **a broad and diverse customer** base.

Number of Transactions

From 1 Jan 2020 to 31 Dec 2021, PT Sejahtera Bersama recorded a total of **3,339 transactions**, reflecting **thousands of orders achieved**, including repeat purchases from existing customers.

Average Sales

The average sales from 1 Jan 2020 to 31 Dec 2021 of **$525.53** indicates that the **revenue per transaction is relatively high**.

Total City

Additionally, from 1 Jan 2020 to 31 Dec 2021, the company successfully reached **361 cities**, indicating a **wide market reach and geographic coverage**.

Quantity Sold

With a total of **11,654 items sold** from 1 Jan 2020 to 31 Dec 2021, the company **offered a wide variety of products**, ranging from items with high revenue to those with lower revenue.

EXPLORATORY DATA ANALYSIS

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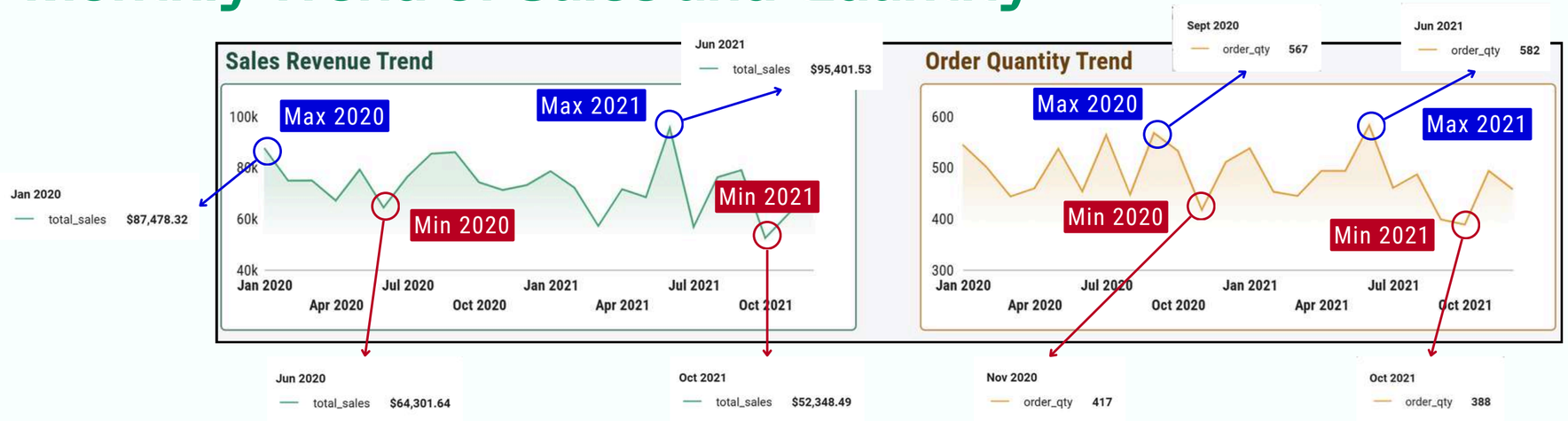
YoY KPI (2021 vs 2020)

Declining Performance

- Total sales declined by 7.5%
- Average sales decreased by 5.2%
- Number of transactions fell by 2.4%
- Quantity sold dropped by 4.5%

All **indicators** are shown with **negative values and red downward** arrows.

Monthly Trend of Sales and Quantity



Fluctuating Performance

The monthly sales and quantity trends **show no** consistent seasonal **pattern**, with **fluctuations occurring across different months**. Peaks and declines appear irregular, indicating **performance is not driven by specific periods**.

In 2021, both **sales revenue and quantity** reached their **peak in June** and hit their **lowest point in October**. In contrast, **in 2020**, the **peak and minimum** months for sales revenue and quantity **occurred in different periods**.

EXPLORATORY DATA ANALYSIS

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Product Category Performance

Revenue & Qty by Product/Category

Selected Column: Category

	Product/Category	Revenue ❶	Qty ❷
1.	Robots	$743,505	1053
2.	Drones	$477,447	1227
3.	Robot Kits	$216,437	1037
4.	Drone Kits	$161,243	1515
5.	Training Videos	$80,716	2081
6.	eBooks	$58,968	3123
7.	Blueprints	$16,435	1618

1 - 7 / 7

Revenue & Qty by Product/Category

Selected Column: Category

	Product/Category	Revenue	Qty
1.	eBooks	$58,968	3123
2.	Training Videos	$80,716	2081
3.	Blueprints	$16,435	1618
4.	Drone Kits	$161,243	1515
5.	Drones	$477,447	1227
6.	Robots	$743,505	1053
7.	Robot Kits	$216,437	1037

1 - 7 / 7

Top Product Category

The **top 5 categories generating the highest profit** are **robot**, **drones**, **robot kits**, **drone kits**, and **training videos**. On the other hand, the **top 5 categories by order quantity** are **eBooks, training videos, blueprints, drone kits, and drone**.

More Value, Higher Profitability

Although **eBooks rank** among the **top** categories **by order quantity**, they **contribute relatively less to total revenue**. In contrast, **products like robots**, despite being **sold** in **smaller quantities**, **generate significantly higher profit**. This indicates that high sales volume does not always lead to high profitability, highlighting **the importance of product value in overall performance**.

Revenue & Qty by Product/Category

Selected Column: Product

	Product/Category	Revenue ❶	Qty ❷
1.	MICR-23K Robot	$157,325	175
2.	RWW-75 Robot	$120,971	137
3.	RLK-9920 Robot	$120,927	173
4.	RQTE-554 Robot	$116,280	170
5.	RCB-889 Robot	$114,192	208
6.	RXW-9807 Robot	$113,810	190
7.	DTI-84 Drone	$105,105	231

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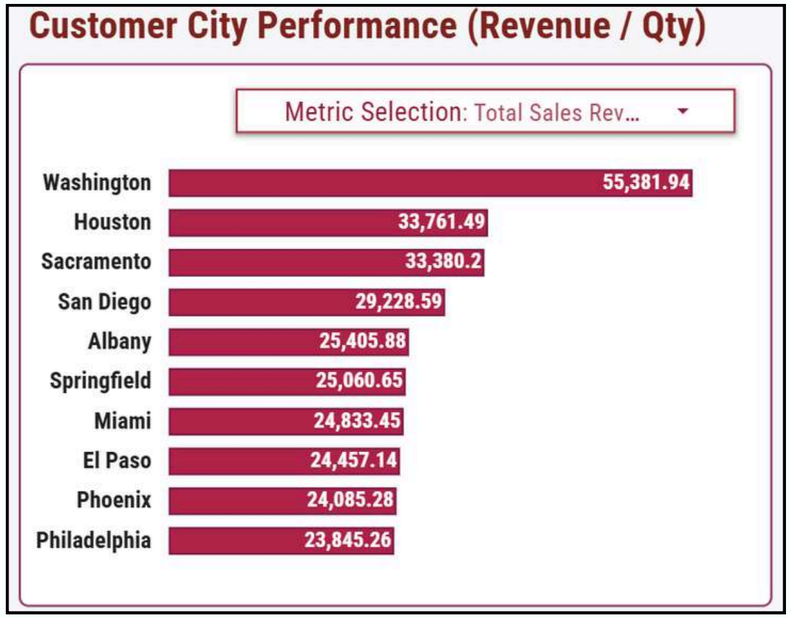
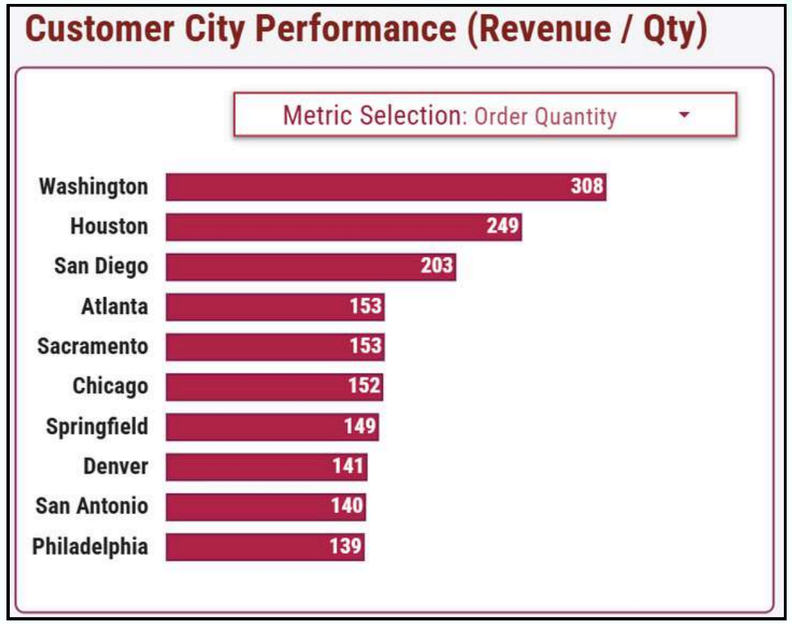
It is shown that when the first column dimension is changed to product name, all top products belong to the robot category.

EXPLORATORY DATA ANALYSIS

Customer City Performance





Top Total Sales Revenue by City

Washington leads as the city with the highest total sales revenue, followed by **Houston and Sacramento**. This indicates that these cities contribute significantly to overall revenue, likely **driven by higher-value transactions**.

Top Order Quantity by City

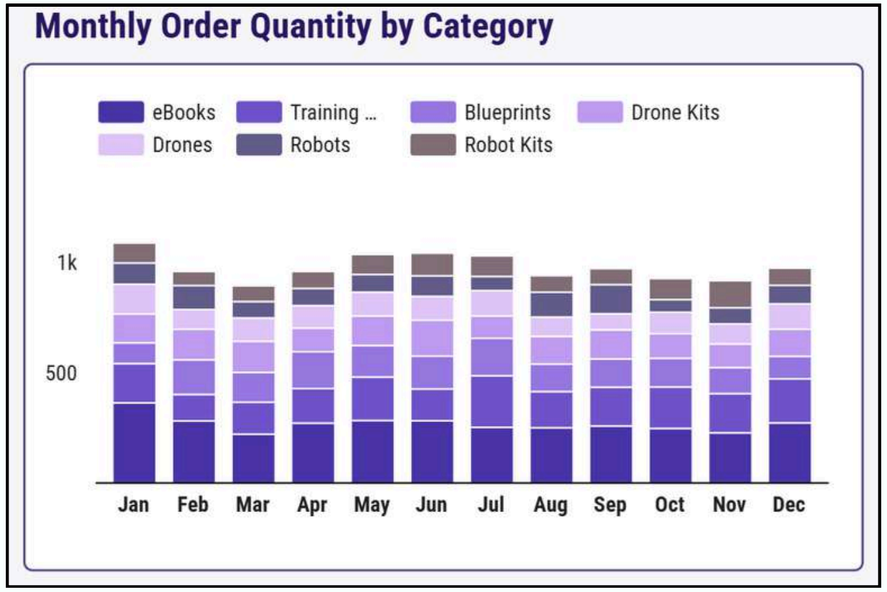
In terms of order quantity, **Washington** also ranks first, followed by **Houston and San Diego**. This shows that these cities have **strong purchasing activity and high customer demand**.

Conclusion

Overall, **Washington** and **Houston** consistently appear in both **revenue and quantity rankings**, indicating they are **key markets with both high demand and strong revenue contribution**. Other cities may excel in only one aspect, suggesting differences between transaction volume and revenue generation across regions.

EXPLORATORY DATA ANALYSIS

Monthly Orders by Category



Fluctuating Demand Across Categories

This chart **reinforces previous findings**, where no clear seasonal pattern is observed across months. It also shows that **eBooks consistently hold the top position** in order quantity. **Training videos** generally **rank second and dominate in most months**, although they are occasionally surpassed by other categories such as blueprints and drone kits. This indicates that **eBooks have the most stable demand**, while other categories show slight variations in their rankings over time.

Repeat Orders

Customer Purchase Summary

	Customer Email	City	Revenue	Purchase Count
1.	hfaulconerbv@msu.edu	Galveston	$9,666	6
2.	aduberryoq@hugedomains.com	Trenton	$7,670	4
3.	wmacdowalb6@themeforest.net	San Diego	$7,565	3
4.	breubens76@taobao.com	Richmond	$7,361	3
5.	genochef@networkadvertising.org	Zephyrhills	$7,221	5
6.	loreillyk9@noaa.gov	Winter Haven	$7,084	6

1 - 100 / 1671

Presence of Repeat Customers

The table shows that **several customers** have **made repeat purchases**, indicating the **presence of customer retention**. However, **only one customer reached up to 7 transactions**, while **others show lower frequencies**, suggesting that **loyalty exists but remains limited**

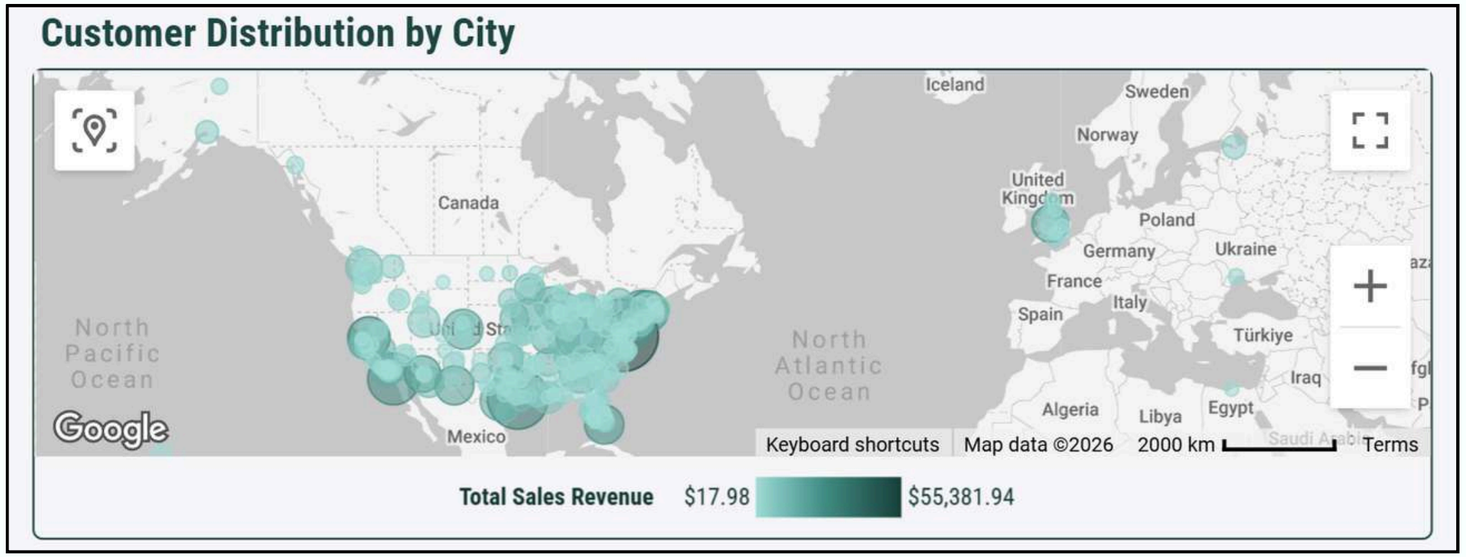
Purchase Count
7
6
6
6
6
6

1 - 100 / 1671

In the long term, **maintaining and strengthening** these **loyal customers is crucial**, as it **helps** ensure **sustainable growth and supports continued business performance**.

EXPLORATORY DATA ANALYSIS

Market Distribution



Current Market Distribution

Market distribution is **highly concentrated in the United States**, which represents **the core market of the company**. In contrast, customer activity in other regions such as the United Kingdom, Europe, and Canada is relatively lower, indicating secondary market presence outside the US.

Untapped Market Potential

There are **significant untapped opportunities** in other regions and continents, as indicated by the relatively sparse distribution of customers on the map. This suggests that **these areas have not yet been fully penetrated** by the company's market activities and **may represent potential growth markets for future expansion**.

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09

KEY INSIGHTS

1 Performance Decline in 2021

In 2021, the company **experienced an overall decline across key KPIs**, with decreases in total sales (-7.5%), average sales (-5.2%), transactions (-2.4%), and quantity sold (-4.5%), indicating **a general slowdown in business performance**.

2 Revenue Driven by High-Value Technology Products

Revenue is **heavily driven by high-value technology products** such as **robots and drones**. Meanwhile, low-price products like **eBooks** contribute **high order volume but generate relatively low revenue**, indicating **strong dependence on premium product categories**.

3 Key Cities Driving Sales Performance

Washington and Houston consistently rank as the top 2 cities in both revenue and quantity, making them **the main contributors to overall sales performance**. Other cities show varying rankings across metrics, indicating less consistent contribution.

4 Market Concentration in the United States

The market is highly concentrated in the **United States as the core demand region**. **Other areas** such as the UK, Europe, and Canada show significantly lower customer activity, **indicating secondary market presence**.

5 Customer Loyalty & Repeat Purchases

Repeat purchases within the same company **reach up to 7 transactions per customer**; however, **the low frequency indicates potential customer churn**, suggesting **weak retention and inconsistent customer return behavior**.

RECOMMENDATIONS

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Product Strategy & Bundling

Strengthen the performance of core products (robots and drones) as the main revenue drivers by ensuring **consistent availability and quality**. Implement **bundling strategies** (robot–robot kit, drone–drone kit) and **thematic packages** (eBooks and training videos) to **increase purchase value**. A **membership program through the official website** can further **enhance customer loyalty while improving profit margins**.

Market Expansion & Distribution

The United States remains the primary market, but sales are still concentrated in a few key cities. **Expanding focus** to **other high-potential cities** can create **a more balanced distribution and maximize market opportunities**. **Enhancing distribution strategies** such as **free shipping** and improving **after-sales services** can strengthen customer experience. At the same time, **the company can gradually explore selected international markets** in line with its **growth readiness and capabilities**.

Customer Retention & Engagement

Improve customer retention through **promotional strategies** (event sales, e-commerce campaigns), **influencer collaborations**, and **user-generated content** to **strengthen awareness and trust**. Introduce **display experiences and interactive events** to allow customers to **directly engage with the products**.

Building **exclusive customer communities**, delivering **high-quality experiences**, and focusing on **small but meaningful service details** can **create strong emotional connections**. **Personalized offers** based on **purchase behavior** can further **drive loyalty and repeat purchases**.

PRESENTATION VIDEO

Youtube Presentation Video Link:

https://youtu.be/bcPcRzwIbk4?si=N_-PM-2_0FTjVwDj

Project Project

THANK YOU

Dashboard Link:

https://datastudio.google.com/reporting/d0ed9cc0-9c9a-4eb8-8d37-d9226df98d5c/page/lZDvF

GitHub Link:

https://github.com/jonathanmarjonoxz/RakaminxMuamalat_BI.git

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